

WIRELESS SENSOR NETWORKS IN SMART AGRICULTURE

A Research Report on WSN-Based Smart Irrigation and Farm Monitoring

Application chosen: Smart Agriculture

Course topic: Wireless Sensor Networks (WSN)

Prepared for: Assignment research project

Prepared by: 00982

This report explains what smart agriculture is, why a Wireless Sensor Network is useful for it, how a WSN-based system can be implemented, and the types of sensors that can be used.

August 2026

1. Introduction

Agriculture is increasingly adopting digital technologies to improve productivity, reduce waste, and manage resources more efficiently. One important technology is the Wireless Sensor Network (WSN). A WSN consists of multiple sensor nodes distributed across an area. Each node senses conditions in its environment, performs basic processing, and communicates data wirelessly to another node, a gateway, or a central system.

In smart agriculture, WSNs can support irrigation management, soil-moisture monitoring, fertilizer optimization, environmental monitoring, and early identification of conditions associated with pests or crop diseases. The central advantage is that information can be collected from many parts of a farm without installing communication cables across the field.

2. What Is Smart Agriculture?

Smart agriculture uses sensors, communication networks, automation, and data analysis to support agricultural decisions. Instead of relying only on occasional manual inspection or fixed schedules, a smart system can monitor conditions continuously and provide information when action is required.

For example, soil in one part of a farm may be dry while another part has sufficient moisture. A WSN can measure both locations and send the results to a central system. The farmer can then irrigate only the dry area, or the system can automatically control the appropriate irrigation equipment.

3. Why a WSN Is Useful for Smart Agriculture

A WSN is useful because agricultural fields can be large, remote, and environmentally variable. Sensor nodes can be placed at different locations to collect local measurements. This allows the farmer to understand differences across the farm and make more precise decisions.

Benefit	Importance in agriculture
Real-time monitoring	Conditions can be measured repeatedly and reported quickly.
Reduced manual inspection	The farmer does not need to physically check every location continuously.
Better water management	Irrigation can be based on measured soil conditions rather than only a fixed schedule.
Scalability	Additional nodes can be deployed to monitor new areas or parameters.
Automation	Sensor data can trigger pumps, valves, alarms, or notifications.

--	--

4. Proposed Application: WSN-Based Smart Irrigation

This report proposes a WSN-based smart irrigation system as a practical smart-agriculture application. The main objective is to monitor soil and environmental conditions and use the collected information to decide when irrigation is required.

4.1 The Problem

Traditional irrigation is sometimes performed according to a fixed timetable. However, crop water requirements can change because of rainfall, temperature, soil type, plant growth stage, and local differences within a field. A fixed schedule may therefore result in unnecessary irrigation or insufficient watering.

4.2 The Proposed Solution

Sensor nodes are distributed across the farm. Each node measures selected conditions, such as soil moisture and temperature. The nodes transmit their measurements through a wireless network to a gateway. The gateway forwards the data to a local server, cloud platform, or farm-management application. The system then compares measurements with decision rules or thresholds and either alerts the farmer or automatically controls irrigation equipment.

5. How the WSN Could Be Implemented

Sensor Layer	Communication Layer	Processing Layer	Action Layer
Soil moisture Temperature Humidity pH Light	LoRa/LoRaWAN Zigbee Wi-Fi NB-IoT	Gateway/Base station Server/Cloud Farm application	Farmer alert Controller Pump Irrigation valve

5.1 Sensor Nodes

Each sensor node can contain one or more sensors, a microcontroller for basic processing, a wireless communication module, and a power source. For remote deployment, batteries combined with solar charging may be considered.

5.2 Wireless Communication

The communication technology should be selected according to farm size, transmission distance, energy consumption, data rate, cost, and available infrastructure. LoRaWAN is often suitable for long-range, low-power sensing, while Wi-Fi may be appropriate where local network infrastructure already exists. Zigbee can support low-power sensor networks over shorter distances.

5.3 Gateway and Data Platform

The gateway receives data from sensor nodes and forwards it to a processing platform. The farmer can view current and historical measurements through a computer or mobile application. The platform may also generate alerts when values exceed acceptable limits.

6. Types of Sensors That Could Be Used

Sensor	What it measures	Possible use
Soil-moisture sensor	Water content in soil	Determine when irrigation is required
Temperature sensor	Air or soil temperature	Monitor crop-growing conditions
Humidity sensor	Moisture in the air	Monitor environmental conditions
Soil pH sensor	Acidity or alkalinity	Support soil-management decisions
Nutrient/EC sensor	Soil nutrients Nitrogen,phosphorus potassium	Support fertilizer and soil monitoring
Light sensor	Light intensity	Monitor sunlight available to plants
Rain sensor	Rainfall or rain occurrence	Avoid unnecessary irrigation
Water-level/flow sensor	Water availability or movement	Monitor tanks, pipes, and irrigation performance

7. System Operation

The proposed system can operate through the following sequence:

Step	Process	Description
1	Sense	Sensors measure soil and environmental conditions.
2	Process	The sensor node performs basic processing and prepares the data for transmission.
3	Transmit	Data is sent wirelessly to a gateway or base station.
4	Analyze	The system stores the data and compares it with rules or thresholds.
5	Decide	The system determines whether an action is required.
6	Act	The farmer is alerted or the irrigation controller activates equipment.
7	Monitor again	The process repeats to provide continuous monitoring.

Sensors → Sensor Nodes → Wireless Communication → Gateway/Base Station → Server/Cloud/Farm Application → Decision → Alert Farmer or Control Irrigation

8. Example Scenario

Consider a farm divided into three sections. Each section contains a sensor node that measures soil moisture. The nodes send readings to the same gateway. The central system can compare the measurements and irrigate only the areas that require water.

Farm section	Soil moisture reading	System decision
Section A	60%	No irrigation required
Section B	25%	Irrigation required
Section C	55%	No irrigation required

In this example, the WSN supports precision agriculture because the decision is based on measurements from individual areas rather than treating the entire farm as if conditions were identical.

9. Advantages of the Proposed System

Advantage	Explanation
Water efficiency	Water can be directed to locations that actually require irrigation.
Continuous monitoring	Measurements can be collected regularly throughout the day.
Early warning	Abnormal readings can trigger notifications.
Reduced labour	Automatic sensing reduces repeated manual inspection.
Data-driven decisions	Historical and real-time measurements support farm management.
Automation	Controllers can operate pumps or valves when predefined conditions occur.

10. Challenges and Limitations

A practical agricultural WSN must also address technical and operational challenges.

- Sensor nodes may have limited battery life, especially when installed far from electricity.
- Large farms can create communication-range problems, and obstacles such as vegetation or terrain can weaken wireless signals.
- Sensors may also be exposed to rain, dust, heat, animals, and agricultural machinery.
- Other important issues include sensor accuracy and calibration, installation and maintenance costs, reliable rural connectivity, network security, and the need to make systems simple enough for effective use in everyday farming.

11. Conclusion

Wireless Sensor Networks provide a practical foundation for smart agriculture because they enable distributed sensing and wireless communication across a farm. By placing sensor nodes in different locations, farmers can obtain information about soil and environmental conditions that may vary across the field.

The proposed smart irrigation system demonstrates the basic WSN process: sensors collect data, sensor nodes transmit the data wirelessly, a gateway forwards the information

for processing, and the system supports a decision or automatic action. Soil-moisture sensing is especially important because it can help determine when irrigation is needed.

Sense → Collect → Transmit → Analyze → Decide → Act.

References

- 1.Hamami, L., & Nassereddine, B. (2020). Application of Wireless Sensor Networks in the Field of Irrigation: A Review. *Computers and Electronics in Agriculture*, 179, 105782.
- 2.Mowla, M. N., Mowla, N., Shah, A. F. M. S., Rabie, K. M., & Shongwe, T. (2023). Internet of Things and Wireless Sensor Networks for Smart Agriculture Applications: A Survey. *IEEE Access*, 11, 145813–145852. <https://doi.org/10.1109/ACCESS.2023.3346299>
- 3.Ojha, T., Misra, S., & Raghuwanshi, N. S. (2015). Wireless Sensor Networks for Agriculture: The State-of-the-Art in Practice and Future Challenges. *Computers and Electronics in Agriculture*, 118, 66–84. <https://doi.org/10.1016/j.compag.2015.08.011>
- 4.Bogena, H. R., Weuthen, A., & Huisman, J. A. (2022). Recent Developments in Wireless Soil Moisture Sensing to Support Scientific Research and Agricultural Management. *Sensors*, 22(24), 9792.